SOFTWARE ENGINEERING

Lab 7: Fuzzing for Input Validation Bugs

Module: Bike Share Rental Parser (`bike_share_rental_parser.py`) | Tool: Hypothesis + pytest

NAME: ANANYA BELIMALLUR RAJASHEKAR

SRN: PES2UG24CS057

SECTION: 5A

LAB07 GITHUB LINK: https://github.com/ananya97br/SE-Lab/tree/main/Lab07

SS1: Buggy module run

`python bike_share_rental_parser.py` on the buggy module. Entering abc crashes with a ValueError traceback.

```
(Lab07) ananya@blaze:~/SE-Lab/Lab07$ python bike_share_rental_parser.py
==== Running parse_ride_duration ====
Enter a ride duration as HH:MM (e.g. 01:15): abc
Traceback (most recent call last):
  File "/home/ananya/SE-Lab/Lab07/bike_share_rental_parser.py", line 39, in <module>
    main()
  File "/home/ananya/SE-Lab/Lab07/bike_share_rental_parser.py", line 28, in main
    print("Duration in minutes:", parse_ride_duration(value))
                                  ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ananya/SE-Lab/Lab07/bike_share_rental_parser.py", line 6, in parse_ride_duration
    hours, minutes = text.strip().split(':')
    ^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
(Lab07) ananya@blaze:~/SE-Lab/Lab07$
```

SS2: Fuzz tests failing on the buggy module

`pytest test_bike_share_rental_parser.py --tb=no`. All 3 Hypothesis tests fail (FFF, 3 failed).

```
(Lab07) ananya@blaze:~/SE-Lab/Lab07$ pytest test_bike_share_rental_parser.py --tb=no
============================= test session starts ==============================
platform linux -- Python 3.12.13, pytest-9.2.0, pluggy-1.7.0
rootdir: /home/ananya/SE-Lab/Lab07
configfile: pyproject.toml
plugins: hypothesis-6.170.14
collected 3 items

test_bike_share_rental_parser.py FFF                                     [100%]

=========================== short test summary info ============================
FAILED test_bike_share_rental_parser.py::test_parse_ride_duration_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_bike_share_rental_parser.py::test_format_station_code_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_bike_share_rental_parser.py::test_parse_ride_log_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================== 3 failed in 0.21s ===============================
(Lab07) ananya@blaze:~/SE-Lab/Lab07$
```

SS2 (details): failing inputs found by Hypothesis

Full `pytest test_bike_share_rental_parser.py` output on the buggy module, in four parts. Shrunk failing inputs: parse_ride_duration: ':', ''', None; format_station_code: ''', None; parse_ride_log: ',', ''', None.

SS2 (details), part 1 of 4

```
(Lab07) ananya@blaze:~/SE-Lab/Lab07$ pytest test_bike_share_rental_parser.py
============================= test session starts ==============================
platform linux -- Python 3.12.13, pytest-9.2.0, pluggy-1.7.0
rootdir: /home/ananya/SE-Lab/Lab07
configfile: pyproject.toml
plugins: hypothesis-6.170.14
collected 3 items

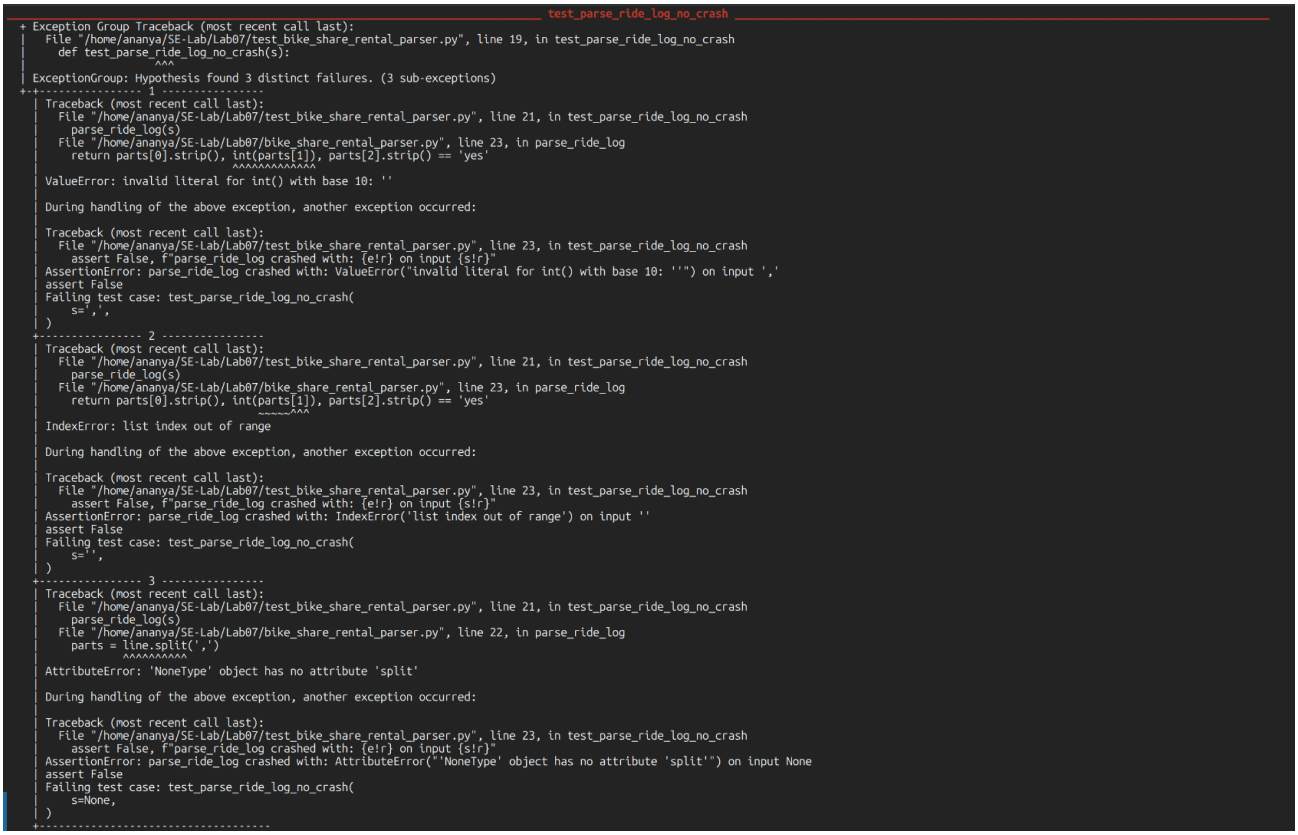
test_bike_share_rental_parser.py FFF                                     [100%]

=================================== FAILURES ===================================
_____________________ test_parse_ride_duration_no_crash ______________________
  + Exception Group Traceback (most recent call last):
  |   File "/home/ananya/SE-Lab/Lab07/test_bike_share_rental_parser.py", line 5, in test_parse_ride_duration_no_crash
  |     def test_parse_ride_duration_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/home/ananya/SE-Lab/Lab07/test_bike_share_rental_parser.py", line 7, in test_parse_ride_duration_no_crash
    |     parse_ride_duration(s)
    |   File "/home/ananya/SE-Lab/Lab07/bike_share_rental_parser.py", line 7, in parse_ride_duration
    |     return int(hours) * 60 + int(minutes)
    |            ^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/ananya/SE-Lab/Lab07/test_bike_share_rental_parser.py", line 9, in test_parse_ride_duration_no_crash
    |     assert False, f"parse_ride_duration crashed with: {e!r} on input {s!r}"
    | AssertionError: parse_ride_duration crashed with: ValueError("invalid literal for int() with base 10: ''") on input ':'
    | assert False
    | Failing test case: test_parse_ride_duration_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/home/ananya/SE-Lab/Lab07/test_bike_share_rental_parser.py", line 7, in test_parse_ride_duration_no_crash
    |     parse_ride_duration(s)
    |   File "/home/ananya/SE-Lab/Lab07/bike_share_rental_parser.py", line 6, in parse_ride_duration
    |     hours, minutes = text.strip().split(':')
    |     ^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/ananya/SE-Lab/Lab07/test_bike_share_rental_parser.py", line 9, in test_parse_ride_duration_no_crash
    |     assert False, f"parse_ride_duration crashed with: {e!r} on input {s!r}"
    | AssertionError: parse_ride_duration crashed with: ValueError('not enough values to unpack (expected 2, got 1)') on input ''
    | assert False
    | Failing test case: test_parse_ride_duration_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/home/ananya/SE-Lab/Lab07/test_bike_share_rental_parser.py", line 7, in test_parse_ride_duration_no_crash
    |     parse_ride_duration(s)
    |   File "/home/ananya/SE-Lab/Lab07/bike_share_rental_parser.py", line 6, in parse_ride_duration
    |     hours, minutes = text.strip().split(':')
    |                      ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
```

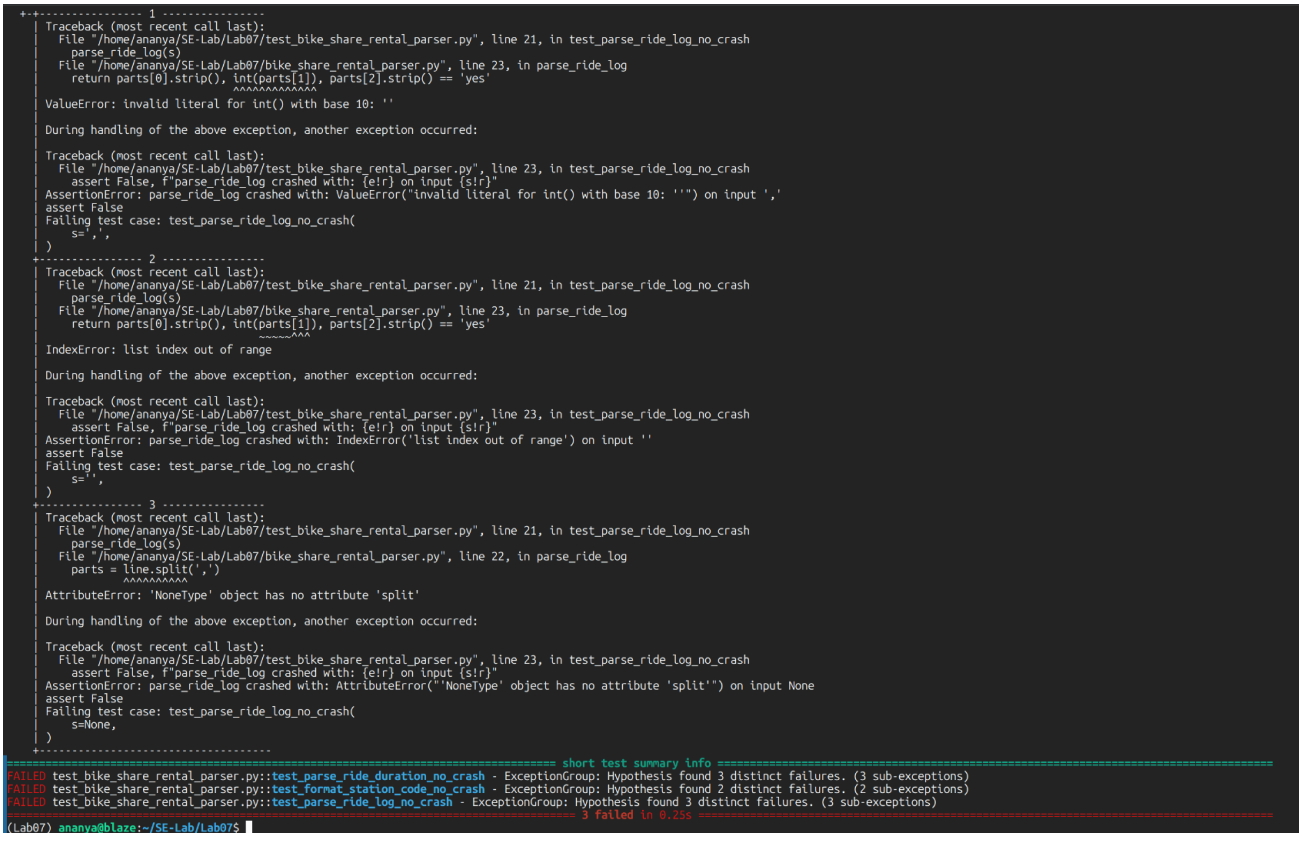
SS2 (details), part 2 of 4

```
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/ananya/SE-Lab/Lab07/test_bike_share_rental_parser.py", line 9, in test_parse_ride_duration_no_crash
    |     assert False, f"parse_ride_duration crashed with: {e!r} on input {s!r}"
    | AssertionError: parse_ride_duration crashed with: AttributeError('"NoneType" object has no attribute \'strip\'') on input None
    | assert False
    | Failing test case: test_parse_ride_duration_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________ test_format_station_code_no_crash ______________________
  + Exception Group Traceback (most recent call last):
  |   File "/home/ananya/SE-Lab/Lab07/test_bike_share_rental_parser.py", line 12, in test_format_station_code_no_crash
  |     def test_format_station_code_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/home/ananya/SE-Lab/Lab07/test_bike_share_rental_parser.py", line 14, in test_format_station_code_no_crash
    |     format_station_code(s)
    |   File "/home/ananya/SE-Lab/Lab07/bike_share_rental_parser.py", line 15, in format_station_code
    |     return code[0] + '-' + code[1:]
    |            ~~~~^^^
    | IndexError: string index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/ananya/SE-Lab/Lab07/test_bike_share_rental_parser.py", line 16, in test_format_station_code_no_crash
    |     assert False, f"format_station_code crashed with: {e!r} on input {s!r}"
    | AssertionError: format_station_code crashed with: IndexError('string index out of range') on input ''
    | assert False
    | Failing test case: test_format_station_code_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/home/ananya/SE-Lab/Lab07/test_bike_share_rental_parser.py", line 14, in test_format_station_code_no_crash
    |     format_station_code(s)
    |   File "/home/ananya/SE-Lab/Lab07/bike_share_rental_parser.py", line 14, in format_station_code
    |     code = code.strip().upper()
    |            ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/ananya/SE-Lab/Lab07/test_bike_share_rental_parser.py", line 16, in test_format_station_code_no_crash
    |     assert False, f"format_station_code crashed with: {e!r} on input {s!r}"
    | AssertionError: format_station_code crashed with: AttributeError('"NoneType" object has no attribute \'strip\'') on input None
    | assert False
    | Failing test case: test_format_station_code_no_crash(
    |     s=None,
    | )
    +------------------------------------
```

SS2 (details), part 3 of 4



SS2 (details), part 4 of 4



SS3: Fixed module run

`python bike_share_rental_parser.py` on the fixed module with the same bad inputs. Each case prints a clean message and no traceback.

```
(Lab07) ananya@blaze:~/SE-Lab/Lab07$ python bike_share_rental_parser.py
==== Running parse_ride_duration ====
Enter a ride duration as HH:MM (e.g. 01:15): abc
Invalid input: expected HH:MM (e.g. 01:15)

==== Running format_station_code ====
Enter a station code (e.g. a12):
Invalid input: station code needs at least 2 characters

==== Running parse_ride_log ====
Enter a ride log line as type,minutes,subscriber (e.g. electric,45,yes): yes
Invalid input: expected type,minutes,subscriber (e.g. electric,45,yes)
(Lab07) ananya@blaze:~/SE-Lab/Lab07$
```

SS4: Fuzz tests passing on the fixed module

`pytest test_bike_share_rental_parser.py` on the fixed module. All 3 tests pass.

```
(Lab07) ananya@blaze:~/SE-Lab/Lab07$ pytest test_bike_share_rental_parser.py
============================= test session starts ==============================
platform linux -- Python 3.12.13, pytest-9.2.0, pluggy-1.7.0
rootdir: /home/ananya/SE-Lab/Lab07
configfile: pyproject.toml
plugins: hypothesis-6.170.14
collected 3 items

test_bike_share_rental_parser.py ...                                     [100%]

============================== 3 passed in 0.29s ===============================
(Lab07) ananya@blaze:~/SE-Lab/Lab07$
```

Bugs found and fixes

Function	Failing input	Error (buggy version)	Fix
parse_ride_duration	None	AttributeError: no .strip on None	isinstance(text, str) check
	" (no colon)	ValueError: not enough values to unpack	Check split(':') gives exactly 2 parts
	':'	ValueError: int('')	isdecimal() on hours and minutes; minutes < 60
format_station_code	None	AttributeError: no .strip on None	isinstance(code, str) check
	"	IndexError: code[0] on empty string	Require at least 2 characters
parse_ride_log	None	AttributeError: no .split on None	isinstance(line, str) check
	"	IndexError: parts[1] missing	Require exactly 3 comma-separated fields
	','	ValueError: int('')	Minutes must pass isdecimal(); bike type non-empty

Reflections:

1. What is fuzzing, and why did it find bugs that normal unit tests might miss?

Fuzzing feeds a program large numbers of random, unexpected or malformed inputs to see whether it crashes. Hypothesis generated inputs such as None, the empty string, ':' and ',' that a person writing unit tests by hand would probably not think of. The original code assumed every input was well formed, so each of these crashed it.

2. What kinds of bugs did fuzzing uncover in this module?

All were input validation bugs: wrong types (None causing AttributeError), missing or empty parts (IndexError and ValueError on unpacking), and non-numeric or empty text passed to int() (ValueError). Each function made an assumption in its docstring that was never checked.

3. How did you fix the code, and how do you know the fix works?

I added defensive checks before using the input: type checks with isinstance, checks on the number of parts after splitting, isdecimal() before int(), and minimum-length and range checks. Invalid input now returns None and main() prints a friendly message. I know it works because the same Hypothesis tests that failed in SS2 now pass in SS4, and the program no longer shows a traceback in SS3.